HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · OIL & GAS · LPG PLANT OPERATIONS

LPG BOTTLING PLANTS · RECEIPT TO DISPATCH · 48 SOPS

Dynamic SOPs, as living diagrams

GenAI turned 20–40 page PDF procedures into flowcharts, sequence diagrams, Gantt plans and mind maps that play step by step, change with the situation and update the day a procedure changes.

~~3wk~~ → **2** days to put an approved SOP change live



48 SOPs
Converted to living diagrams
receipt, filling, dispatch, safety



2 days
Time to update an SOP on the floor
▼ from about 3 weeks



−42 %
SOP-related audit observations
▼ year on year, pilot plants

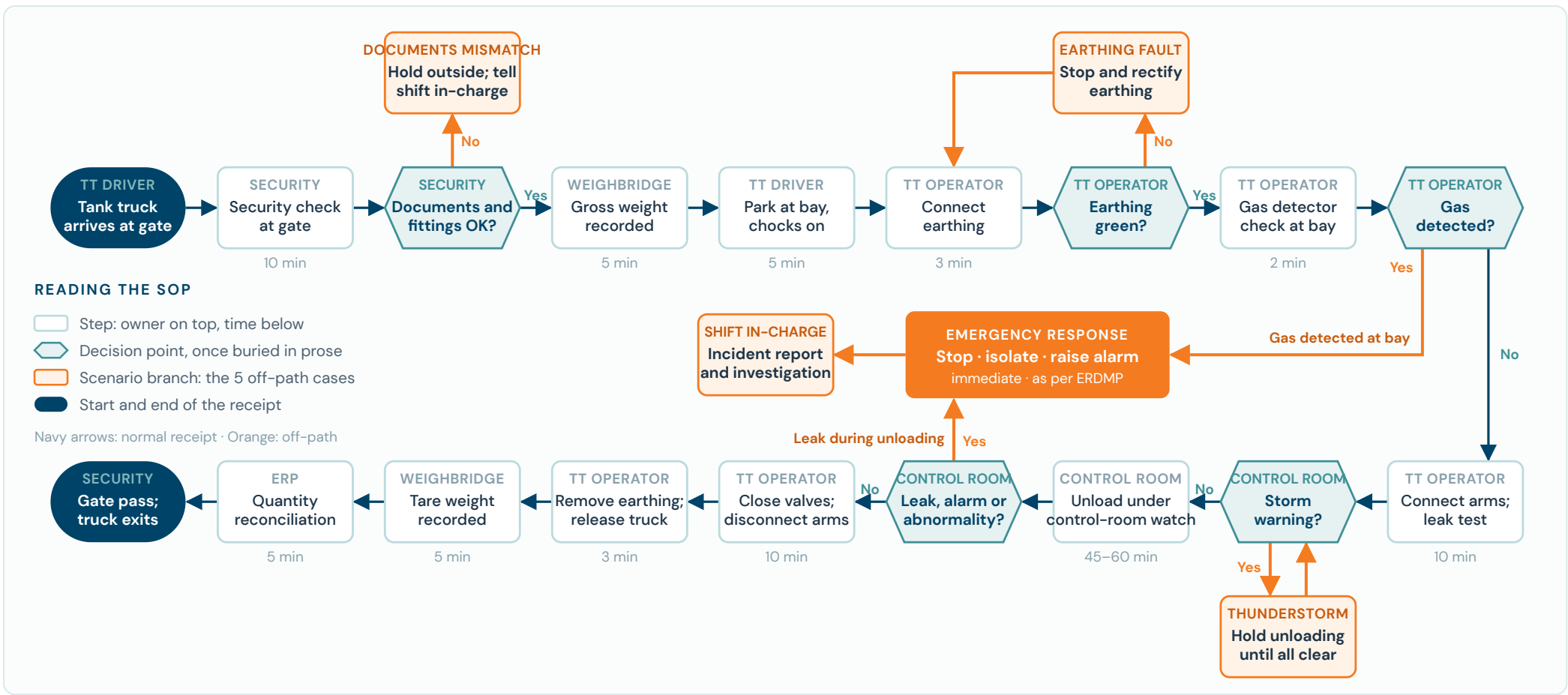


6 diagram types
Views from one source of truth
one text file, many formats

Rings: one segment per SOP converted; 2 days against about 3 weeks (inner ring = before); observations remaining against last year (inner = 100%); one segment per diagram type. Figures indicative; validated metrics available on request.

01 One SOP, every branch visible

Bulk LPG tanker receipt as a dynamic flowchart · procedure illustrative and simplified



Live SOPs are owned and approved by the plant and follow its management-of-change process and applicable OISD and PESO requirements. The night-shift buddy step (v2.2) appears only on night shifts and is not drawn here.

02 One text file, six pictures

Each diagram answers a different question on the floor · content illustrative

THE SOURCE · MERMAID

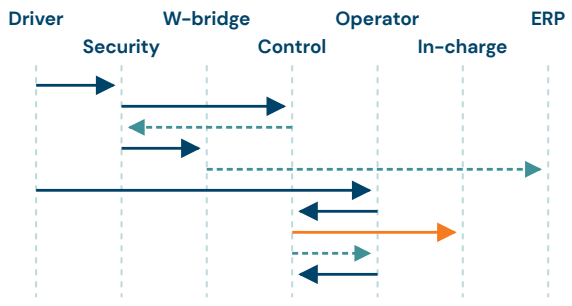
Written once, as code

```
flowchart TD
  GATE(["Tank truck arrives at gate"])
  SEC["Security check at gate"]
  DOCOK{"Docs and fittings OK?"}
  EARTH["Connect earthing"]
  WX{"Storm warning?"}   %% v2.1
  GATE --> SEC --> DOCOK
  DOCOK -- "No" --> REJ
  DOCOK -- "Yes" --> WB1 --> BAY
  GASQ -- "Yes" --> ERS
```

Plant SMEs own the text; diagrams, playback and versions follow automatically.

SEQUENCE

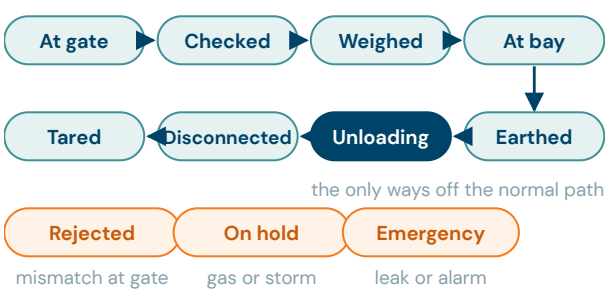
Who tells whom, and when?



Hand-offs between seven roles; orange is the emergency call to the shift in-charge.

STATE

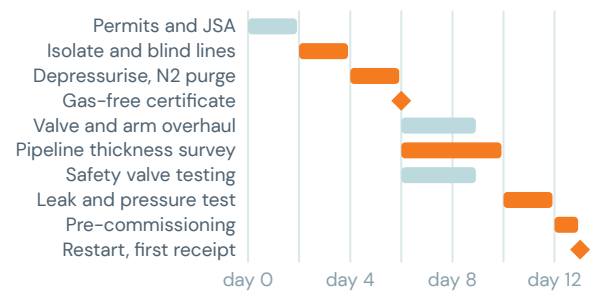
Where is this tanker now?



Every tanker is always in exactly one state; the control room sees each truck on this map.

GANTT

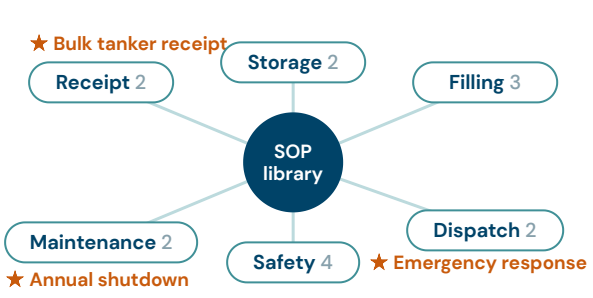
When will the shutdown finish?



Annual shutdown plan; orange is the critical path. Repair crews in parallel: 13 days.

MIND MAP

Which SOP do I need?



The plant's SOP library at a glance; ★ marks SOPs already live as dynamic SOPs.

TIMELINE

What changed, and why?



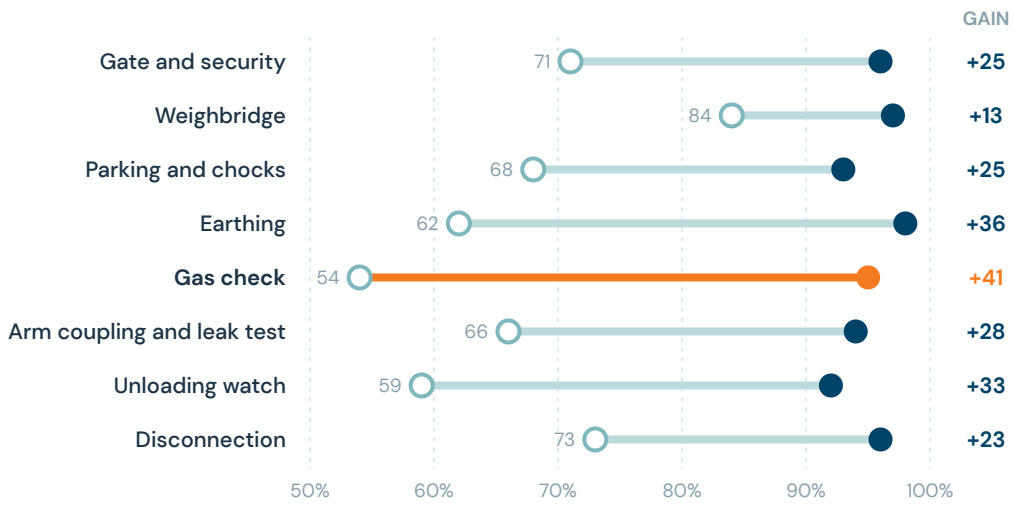
Every change versioned with its reason, so auditors see what changed and why.

03 Followed, not just filed

Impact after go-live · figures indicative

Audit compliance by SOP step

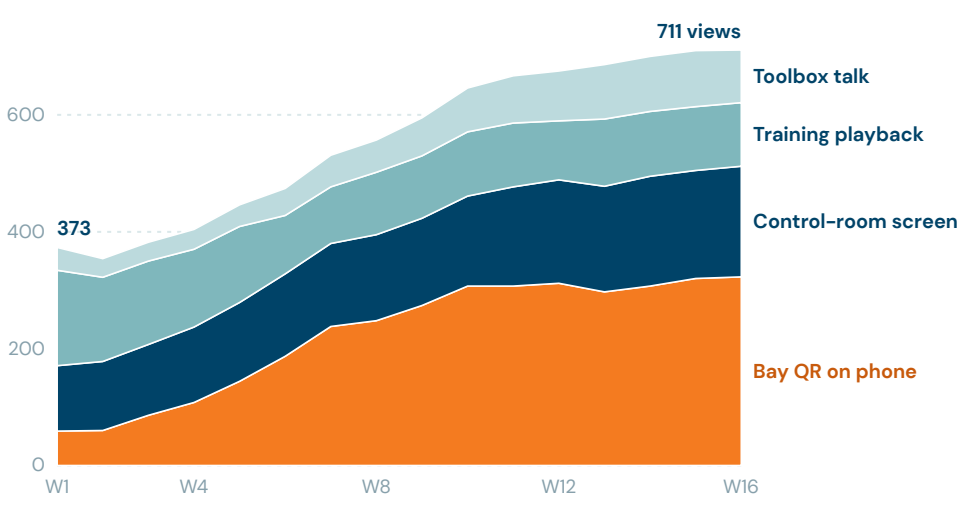
Bulk tanker receipt, % of audited receipts done fully to SOP · ○ PDF SOP ● Dynamic SOP



67% → **95%** average across the 8 audited steps · axis starts at 50%

Where the SOPs get opened

Weekly SOP views by channel, first 16 weeks after go-live (stacked)



Bay QR views: **59** → **323** a week; training playback settles as inductions end

04 Four pain points, four fixes

Before → after, how an SOP lives at the plant

BEFORE
Long PDFs
20 to 40 pages of text. Operators knew the gist; few could find the exact step under pressure.

SIX VIEWS, ONE FILE

AFTER
Pictures that play
Flowchart, sequence, state, Gantt, mind map and timeline from one file.

BEFORE
Branches buried in prose
"If gas is detected, then…" sat in paragraph six; hand-offs were implied in the text.

SCENARIO PLAYBACK

AFTER
Visible branches
Decision points drawn out; colour-coded owners, filter to "my steps".

BEFORE
Slow to change
After a near-miss or a change approval, edit, approve and re-print took about three weeks.

CHANGE ONCE

AFTER
Live in about 2 days
Approved change merged into the file; every view updates, with history.

BEFORE
Old copies in the field
Laminated copies at bays and in contractor packs drifted out of date; audits flagged it.

QR AT THE BAY

AFTER
Always current
The bay QR opens the current version; inductions use step-by-step playback.

05 How HMC ran it: GenAI drafts, experts approve

Five phases · deliverable at the end of each

2 WEEKS
Discover
Walked the plant, SOP in hand, with operators and safety officers. Inventoried 48 operating and safety SOPs; marked where people hesitated.
Deliverable · **SOP inventory, prioritised by risk and use**

2 WEEKS
Design
The right picture per question: flowchart for "what next?", sequence for "who tells whom?", Gantt for shutdowns.
Deliverable · **Diagram standard, role colours, naming rules**

4 WEEKS
Build
Claude converted each PDF into Mermaid code; every step tagged with owner, time, PPE, checks and source clause. Plant SMEs reviewed every diagram.
Deliverable · **SOP studio: playback, scenarios, role filters**

2 MONTHS
Adopt
QR codes at bays open the live flowchart; the control room plays the sequence view; inductions use playback, not slides.
Deliverable · **Bay QR rollout and training playbook**

∞ ONGOING
Scale
Management of change ends with a code edit, not a reprint; version timeline kept for audit; template extended to other plants and terminals.
Deliverable · **SOP-as-code governance and rollout plan**

What transfers to any operations team

01
Write the SOP once, as code
A short text file can become every diagram a team needs. Change the file and every view follows.

02
Show the branches
Most incidents happen off the happy path. Making "what if" visible, and playable, is where the value is.

03
GenAI drafts, experts own
Conversion is fast with AI. Trust comes from plant experts approving every step and every change.

Built with **Claude** · **Mermaid.js** · **D3.js** · **Version control** · **QR codes** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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